

Practical Demo: Vasicek Rate Scenarios and Long-Run Behavior

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Purpose and Assumptions

The Vasicek model is

$$dr_t = \kappa(\theta - r_t)dt + \sigma dW_t.$$

This demo connects its parameters to a compact rate history, simulates scenarios, checks long-run moments, and compares model-based changes with a simple historical shock. Rates and parameters are **illustrative**, not a current calibration. The short rate is represented by a monthly overnight-rate proxy; parameters are constant; estimation error, regime shifts, jumps, and term premiums are ignored.

1 Compact Illustrative Rate History

```
import numpy as np
import pandas as pd
import matplotlib.pyplot as plt

dates = pd.date_range("2023-01-31", periods=30, freq="ME")
rates = np.array([
    0.0430, 0.0450, 0.0470, 0.0490, 0.0510, 0.0525,
    0.0530, 0.0535, 0.0532, 0.0528, 0.0525, 0.0520,
    0.0515, 0.0508, 0.0500, 0.0492, 0.0484, 0.0475,
    0.0468, 0.0460, 0.0455, 0.0450, 0.0446, 0.0442,
    0.0439, 0.0436, 0.0434, 0.0432, 0.0431, 0.0430,
])
history = pd.Series(rates, index=dates, name="Illustrative short rate")
history.to_frame().tail()
```

Illustrative short rate	
2025-02-28	0.0436
2025-03-31	0.0434
2025-04-30	0.0432
2025-05-31	0.0431
2025-06-30	0.0430

The series shows a tightening phase followed by gradual normalization. It is too short for a robust statistical calibration, so the demo uses transparent, economically plausible parameters informed by its scale rather than presenting false precision.

2 Parameters and Financial Meaning

```
r0 = float(history.iloc[-1])
kappa = 0.70      # annual speed of mean reversion
theta = 0.0380    # long-run mean rate
sigma = 0.0180    # annualized short-rate volatility
dt = 1 / 12

half_life = np.log(2) / kappa
```

```

long_run_std = sigma / np.sqrt(2 * kappa)

pd.Series({
    "Current rate": r0,
    "Mean-reversion speed kappa (1/year)": kappa,
    "Long-run mean theta": theta,
    "Volatility sigma (rate/sqrt(year))": sigma,
    "Mean-reversion half-life (years)": half_life,
    "Long-run standard deviation": long_run_std,
}).round(5)

```

```

Current rate                0.0430
Mean-reversion speed kappa (1/year)  0.7000
Long-run mean theta         0.0380
Volatility sigma (rate/sqrt(year))  0.0180
Mean-reversion half-life (years)    0.9902
Long-run standard deviation  0.0152
dtype: float64

```

- κ controls how quickly shocks decay; the half-life is $\ln(2)/\kappa$.
- θ is the long-run mean under the chosen probability measure.
- σ controls the size of random innovations and the long-run dispersion.

3 Simulate Monthly Paths

The exact Vasicek transition is normal:

$$r_{t+\Delta} = \theta + (r_t - \theta)e^{-\kappa\Delta} + \sigma\sqrt{\frac{1 - e^{-2\kappa\Delta}}{2\kappa}}Z.$$

```

rng = np.random.default_rng(411)

def simulate_vasicek(initial_rate, years, paths, rng):
    steps = int(round(years / dt))
    output = np.empty((steps + 1, paths))
    output[0] = initial_rate
    persistence = np.exp(-kappa * dt)
    innovation_std = sigma * np.sqrt((1 - np.exp(-2 * kappa * dt)) / (2 * kappa))
    for t in range(steps):

```

```

        output[t + 1] = (
            theta + (output[t] - theta) * persistence
            + innovation_std * rng.standard_normal(paths)
        )
    return output

```

```

paths = simulate_vasicek(r0, years=5, paths=5_000, rng=rng)
time = np.arange(paths.shape[0]) * dt

```

```

fig, ax = plt.subplots(figsize=(7.5, 4.1))
for i in range(30):
    ax.plot(time, 100 * paths[:, i], alpha=0.28, linewidth=0.8)
ax.plot(time, 100 * paths.mean(axis=1), color="black", linewidth=2.2, label="Simulation mean")
ax.axhline(100 * theta, color="#F58518", linestyle="--", label="Long-run mean")
ax.set(xlabel="Years", ylabel="Short rate (%)", title="Illustrative Vasicek scenarios")
ax.grid(alpha=0.3)
ax.legend()
plt.tight_layout()
plt.show()

```

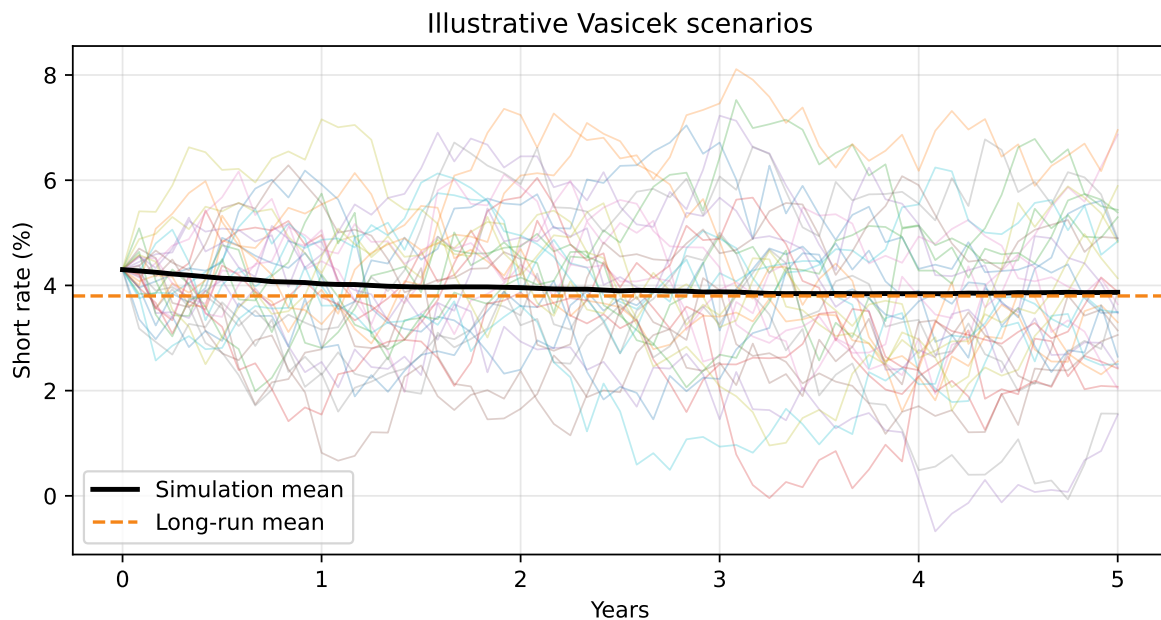


Figure 1: Vasicek paths fluctuate randomly but revert toward the long-run mean

4 Distribution and Negative Rates

```
horizon_rate = paths[-1]
summary = pd.Series({
    "Five-year mean": horizon_rate.mean(),
    "Five-year standard deviation": horizon_rate.std(ddof=1),
    "5th percentile": np.quantile(horizon_rate, 0.05),
    "Median": np.median(horizon_rate),
    "95th percentile": np.quantile(horizon_rate, 0.95),
    "Probability of negative rate": np.mean(horizon_rate < 0),
})
summary.round(5)
```

Five-year mean	0.0387
Five-year standard deviation	0.0150
5th percentile	0.0135
Median	0.0387
95th percentile	0.0634
Probability of negative rate	0.0064
dtype:	float64

Vasicek rates are normally distributed, so negative rates are possible. That can be useful in markets where negative policy rates are plausible, but it may be undesirable for other applications. A positivity-preserving model such as CIR changes this feature, at the cost of a different distribution and dynamics.

5 Model Scenario Versus a Historical Shock

Use the largest one-month decline in the compact history as a simple historical shock, and compare it with the model's one-month distribution.

```
monthly_changes = history.diff().dropna()
historical_down_shock = monthly_changes.min()
model_one_month = simulate_vasicek(r0, years=dt, paths=50_000,
                                   rng=np.random.default_rng(412))[-1] - r0

pd.Series({
    "Largest historical monthly decline (bp)": historical_down_shock * 10_000,
```

```

    "Model mean one-month change (bp)": model_one_month.mean() * 10_000,
    "Model 5th-percentile change (bp)": np.quantile(model_one_month, 0.05) * 10_000,
    "Model 1st-percentile change (bp)": np.quantile(model_one_month, 0.01) * 10_000,
}).round(2)

```

```

Largest historical monthly decline (bp)    -9.00
Model mean one-month change (bp)          -2.67
Model 5th-percentile change (bp)          -86.08
Model 1st-percentile change (bp)          -118.84
dtype: float64

```

The historical scenario preserves an observed move but supplies only one outcome. The Vasicek simulation supplies a distribution and mean reversion, but depends on model specification and parameters. Risk management should compare both rather than treating either as truth.

6 Validate Long-Run Moments and Units

The stationary Vasicek distribution has

$$E[r] = \theta, \quad \text{Var}(r) = \frac{\sigma^2}{2\kappa}.$$

```

long_paths = simulate_vasicek(r0, years=20, paths=30_000,
                             rng=np.random.default_rng(413))[-1]
simulated_mean = long_paths.mean()
simulated_std = long_paths.std(ddof=1)

assert kappa > 0 and sigma > 0
assert abs(simulated_mean - theta) < 0.0005
assert abs(simulated_std - long_run_std) < 0.0005
assert abs((0.0001 * 10_000) - 1.0) < 1e-12

pd.Series({
    "Theoretical long-run mean": round(theta, 5),
    "Simulated 20-year mean": round(simulated_mean, 5),
    "Theoretical long-run standard deviation": round(long_run_std, 5),
    "Simulated 20-year standard deviation": round(simulated_std, 5),
    "Decimal 0.0001 equals one basis point": True,
})

```

Theoretical long-run mean	0.038
Simulated 20-year mean	0.0382
Theoretical long-run standard deviation	0.0152
Simulated 20-year standard deviation	0.0154
Decimal 0.0001 equals one basis point	True
dtype: object	

7 Financial Interpretation

Mean reversion makes the expected rate move toward θ , while volatility keeps future rates dispersed. κ determines how quickly today's rate loses influence; σ determines scenario width. The model is a disciplined scenario generator, not a forecast guarantee. Calibration window, probability measure, structural breaks, and the normal model's negative-rate tail all affect how its output should be used.